

Test 1A Review Questions

Student edition

These are questions worth focusing on for the test. Use them to guide your review of Modules 1A and 2A, the assigned videos, and the Retro and Cosmic Redshift slide quizzes. Brief answers are enough for most prompts; practice explaining the ideas in your own words.

The Universe and Cosmic Beginnings

1. What is the Universe, and what does cosmology study?

2. What is meant by a multiverse, and how does that idea relate to physical constants that permit life?

3. Approximately how old is the Universe, and how does its expansion help us reconstruct its past?

4. How does the raisin-cake analogy explain expansion without a special center or surrounding empty space?

The Early Universe and the Cosmic Microwave Background

5. Which light elements formed during primordial nucleosynthesis in the first 20 minutes?

6. When was the cosmic microwave background released, and what change in matter allowed light to travel freely?

7. Who discovered the cosmic microwave background, and why was the discovery unexpected?

8. What is the CMB's temperature today, and why has it cooled since its release?

9. How does inflation help explain the CMB's nearly uniform temperature and the seeds of galaxies?

Dark Matter and Dark Energy

10. Approximately what percentages of the Universe are ordinary matter, dark matter, and dark energy, according to 1AE2?

11. How do galaxy rotation measurements provide evidence for dark matter?

12. What did observations of Type Ia supernovae reveal about cosmic expansion?

13. What is dark energy, and what is the Big Chill scenario for the Universe's future?

Olbers and Cosmic Redshift

14. What is Olbers' paradox, and which assumptions about the Universe produce it?

15. How do the Universe's finite age and the redshifting of light help explain the dark night sky?

16. How can light from a distant star reach Earth billions of years after the star has died?

Aristotle and the Experimental Method

17. What did Aristotle mean by "Knowledge is the product of experience," and how does that connect to scientific evidence?

18. Why is the traditional list of five senses incomplete? Give two additional senses discussed in the videos.

19. How do telescopes and other instruments extend our senses beyond visible light?

20. What is the definition of an experiment? Why is it crucial for an experiment to be repeatable?

21. What is a scientific hypothesis?

22. How does the scientific meaning of “theory” differ from its everyday use to mean a guess?

Physical Laws and Conservation

23. How does a physical law differ from an advisory principle in the scientific method?

24. What is a conserved physical quantity? Give three examples from classical physics.

25. What does Galileo’s law of falling bodies predict for two objects of different masses dropped from the same height when air resistance is negligible?

26. What two objects did Commander Scott drop on the Moon, and how did the demonstration illustrate Galileo’s law?

Occam's Principle and the Principle of Falsifiability

27. When two competing hypotheses fit the same observations, what does Occam's principle advise, and does it guarantee that the simpler hypothesis is correct?

28. What does it mean for a hypothesis to be falsifiable?

Solar System Models and Retrograde Motion

29. What is apparent retrograde motion against the background stars? Does the planet actually reverse its orbit?

30. Why does Mars appear to move backward when Earth overtakes it, and how is this related to opposition?

31. Why does Venus exhibit retrograde motion near inferior conjunction, and which planet overtakes the other?

32. How does the heliocentric explanation of retrograde motion illustrate Occam's principle?

The Zodiac and Cosmic Geometry

33. What is the zodiac, and how many constellations does the Sun actually pass through?

34. What do CMB measurements indicate about the Universe's geometry, and why does near-flatness alone not settle whether its total extent is infinite?

Cosmic Discoveries and Evidence

35. What contributions did Friedmann and Lemaître make to the idea of an expanding Universe?

36. What contributions did Hubble, Slipher, and Humason make to the idea of an expanding Universe?

37. Which two scientists are associated with the discovery of dark matter, and what observations did they use?

38. Is cosmic inflation a confirmed fact or a hypothesis in the episode? What distinction should you make between evidence for it and certainty?

39. Before the 1998 supernova results, why did astronomers expect cosmic expansion to slow down?
